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Voter2 Transition Guide (From Voter/RTCM)

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DRAFT VERSION: 1.00

APRIL 26, 2026

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1 INTRODUCTION

Realizing that almost everyone using a Voter2 will have first had at least some if not extensive experience with the Voter/RTCM already, this is a short guide to help you get started with the Voter2 assuming you are already familiar with the Voter/RTCM.

From here on, I will refer to the Voter/RTCM as just the Voter.

It is important to note that the Voter2 is a ground-up project, not borrowing any code from the Voter (this is explained in the Users Guide). I had very little experience with the Voter before embarking on the Voter2 and even during Voter2 development used it very little, mainly to ensure interoperability, never in an actual "real world" environment. As such, I am aware I have almost certainly made some usage assumptions that are not normal operating practice. I look forward to that feedback so I can make it a better project over time.

2 SCOPE

This is not intended to be a replacement for the Hardware Guide or User Guide documents. They are the reference documents for the Voter2. This is intended to be a cheat sheet for folk already familiar with the Voter to make the transition to the Voter2 a bit easier, highlighting the similarities, differences, and enhancements over the Voter. It just notes these differences to save some flailing and know what to look for in the Hardware and Users guides.

3 INTEROPERABILITY

3.1 PROTOCOL

By requirement, the Voter2 is fully compatible with the Voter protocol as implemented by the ASL1 and ASL3 platforms. The Voter2 must operate with these platforms without any modifications to them. As with any non-trivial protocol there are nuances and boundary cases I've run in to, but whenever there has been an issue or question, the reference is the ASL servers and compatibility with them.

3.2 VOTING

For voting, the Voter2 can fully inter-operate in a network with a mix of Voters and Voter2s.

3.3 SIMULCASTING

For simulcasting, the initial goal is interoperability between Voter2s only. This is in keeping with the general requirement that for simulcasting all hardware (Voters, repeaters, etc.) should be identical. As a stretch goal it would be nice to be able to mix Voters and Voter2s while still keeping the same repeater.

3.4 TBD

4 DIFFERENCES

4.1 USER TYPES

The Voter2 supports two users, `root` and `guest`. Root has full access while guest cannot save changes to NVM, change passwords or do software updates. The intent is letting the guest make almost any changes they like, but a reboot will bring the Voter2 back to the last root-defined configuration.

4.2 COMMAND LINE INTERFACE

The biggest most obvious difference from the Voter is the user interface, transitioning from menu-based to command line based. Core to the Voter2 is future expandability and the menu-based interface doesn't support that gracefully. This unfortunately does mean more typing compared to the Voter. Command and parameter shortcuts hopefully help.

The `help` command lists all commands. `help` followed by a command is more detailed syntax for that command. Finally for commands like `set/show`, `dial`, and `local` that have sub-settings, `help` followed by the command and sub-setting will give the syntax for that sub-setting.

The command prompt starts with an optional device name (see voter name in enhancements), root/guest ID, a dirty flag, and finally a bracket. The dirty flag is set if there have been any settings changes made to memory that have not been saved to NVM. This is your reminder that some setting will get lost if you reboot.

I do want some minimal form of command recall/editing!

4.3 DEBUG LOGGING

Another major difference from the Voter is both how and where debugging logs are handled.

For the Voter2 the debugging output is on a separate interface, not shared with the console. This keeps the console fully usable even if there are considerable (and ongoing) debug messages being displayed. The initial port for debugging messages is the UART on the 16-pin ST debug header. This is the only port available until the RTOS and Ethernet stack are running, so any debug messaging regarding boot and the final flash reprogramming phase of a software update process are here. Once Ethernet is available you can Telnet into port 24 of the Voter2 and then debugging messages will be available there. Once the Telnet session is terminated, debugging messages revert back to the UART.

What debugging logs to show is also different. Instead of single debug level for all messages, debugging can be set by functional block, so just for telnet or GPS for example. Then for each block there are four levels of information, `none`, `major`, `support` and `io` (most verbose). Type in `help set logging` for command details.

4.4 RSSI DETERMINATION

The Voter2 supports two sources of RSSI information. For the Voter2-2K, the primary source is intended to be the analog RSSI signal from the MTR-2000. The alternate source is similar to the Voter in that the energy level above 2.4KHz is measured and is inversely proportional to signal level. This latter method still needs to be refined!

4.5 RECEIVE LEVEL CALIBRATION

Receive audio level setting is very similar to the Voter in that you set up the 1KHz tone with 3KHz deviation on your service monitor, enter the calibration command on the Voter2 (`status level` for the Voter2), and then adjust.

The big difference for the Voter2-2K is that the analog level adjustment is done on the MTR-2000, not the Voter2! See Appendix C of the Users guide for details.

The other difference is the Voter2 has two level bars, the one on the left is the course level setting to get you close, the second is "zoomed in" on the target level to make tweaking a bit easier visually.

4.6 SOFTWARE UPDATES

The Voter2 relies on TFTP for doing software updates. This may be a bit more cumbersome for a single stand-alone setup, but optimized for a networked environment where one central TFTP server can support all Voter2s on the network and the updates can be performed remotely (use with care!). The `set tftpaddr` command sets the address of the TFTP server and software updates are a three-step process.

- `Update check`: shows the current software version and the one on the server
- `Update download`: downloads the update to a staging NVM on the voter2
- `Update program`: programs the executable NVM from the staging NVM

The staging NVM step allows for a complete download and verification of the image before programming the executable NVM. This way the most vulnerable part of the update process—programming the executable flash—needs only stable power to complete.

On my to do list is the ability to update via the console for non-networked setups.

4.7 SAMPLE SYNCHONIZATION

This is getting down into nuances of the protocol, but could be important for any strange artifacts with voting. Core to keeping multiple voters synchronized is the voters sending EXACTLY 8,000 samples per second—GPS-enforced using 1PPS as a reference.

For the Voter, the clock running the A/D is running from a local crystal whose frequency will be off and indeed may drift over time/temperature/etc. The Voter counts samples every GPS 1PPS and if the total number of samples is off by 1 or 2, either adds or drops a sample to get back to exactly 8,000.

For the Voter2 there is a PLL (a combo timer/software one) that generates an 8KHz sample clock that is phase and frequency locked to the GPS 1PPS, so there is never any drifting. The net result is the same, averaging exactly 8000 samples per second, but implemented very differently. They differ on instantaneous jitter. The Voter jitter is ±1-2 samples (250us) while the Voter2 is ±4 internal clocks or 400ns.

4.8 TBD

5 ENHANCEMENTS

5.1 REMOTE UART

The Voter2 supports the RFC2217 (COM port control) standard, allowing remote com port operations over Ethernet. This feature along with an RFC2217 driver on your computer allows you to run the Motorola RSS software and access/control the MTR-2000 remotely. See Appendix A for details.

5.2 TEMPERATURE SENSORS

The Voter2 has two on-board temperature sensors, one internal to the MCU which is useful for ensuring the CPU isn't getting too hot (VERY UNLIKELY at 125C) and the second one external to the MCU for ambient temperature. There is support for additional sensors off-board too.

5.3 ADDITIONAL DEBUGGING FEATURES

In addition to the `set logging` command, several other debugging features exist. These have been useful to me for developing the Voter2 software, but I have deemed useful enough to retain going forward. This is a listing of the commands, see the Users guide for more details

- `netdelay` – histogram of packet delivery times to show network jitter
- `rxtest` – sends audio packets with a test pattern for sync calibration
- `txtone` – Turns on a test tone with programmable frequency and amplitude
- `set flags, set dlogger, set errlogs` – These are very "down and dirty" debugging commands only useful for development work, but if a developer is helping you with an issue, likely you will be asked to use them.

5.4 BACKUP/RESTORING SETTINGS

The Voter2 has support for saving settings to a server (`settings backup`) and restoring said settings from a server (`settings restore`). This uses the same TFTP server address as used for software downloads. The settings are saved as a series of `set` commands, so you can manually copy and edit a settings file.

5.5 EXPANDABILITY

With the Voter2 nearly feature complete, only 5% of the flash is being used. There is LOTS of room for future enhancements. CPU cycles and RAM are similarly unburdened.

5.6 GPS SETTINGS

This is a minor usability upgrade. For the supported GPS modules, instead of looking up the baud rate and active edge, the `set gpsdev` command knows the settings for various GPS receivers and sets the parameters for you. Right now, the only two supported GPS receivers are the ST Teso LIVF4 and Ublox M8N modules, but this command can be extended as more GPS receivers get support. For any one-off GPS receivers, commands for baud rate and active edge can still be set manually.

5.7 ANALOG PROBE

The advantage of doing audio processing with DSP is its flexibility. The downside is visibility, you can't normally probe the signal with your oscilloscope as it passes through DSP routines. This command addresses that issue. It uses the second DAC port to show the signal between various filters in the Rx and Tx DSP chains. You put your scope probe on the ANLG pin of the debug header (P5) and use the `set probe` command to "probe" different points of the internal DSP chain.

5.8 VOTER NAME

Another usability upgrade idea from Dave (WA1JHK). You can assign a name to your Voter2 that becomes part of the command prompt. `set myname` Handy when you are dealing with multiple Voter2s on the same network and can't keep track of which one you are dealing with.

5.9 TBD

6 VOTER-VOTER2 COMMAND CHEAT SHEET

This is just a quick mapping of Voter menu commands to Voter2 CLI commands with the shortcut in parenthesis. This is just the command mapping, the syntax is often different. Major differences are noted.

6.1 Main Menu

Voter	Voter2	Note
1	Show myserialnum (msn)	Set only works once! (value put in OTP)
2	Set dial fqdn (qdn)	
3	Set dial port (prt)	
4	N/A	No local port override support
5	Set dial voterpwd (vpw)	
6	Set dial hostpwd (hpw)	
7	Set dial txdly (txd)	Voter2 parameter is in decimal ms, down to ns. Combines Voter 7 and 19 commands into one.
8-9	N/A	Currently NMEA support only, no inversion support yet
10	Set gpsedge (ged)	Set gpsdev for device lookup
11	Set gpsbaud (gbd)	Set gpsdev for device lookup
12	Set rxmode (rxm)	More parameters in Voter2
13	Set rxmode (rxm)	More parameters in Voter2
14	Set logging (log)	More parameters in Voter2
15-16	N/A	No Alternate server support yet
17	N/A	Not in Voter either
18	Set local hangtime (htm)	I think this is the equivalent?
19	Set dial txdly (txd)	Combined with command #7
97	Status level (lvl)	
98	Status xxx	Many status commands
99	Settings save	
I	See IP Menu	
O	See Online Menu	
S	N/A	May need something here
Q	Logout (exit)	
R	Reboot	
D	N/A	Scattered in other Voter2 commands

6.2 IP Menu

Voter	Voter2	Note
1	Set mystaticip (msi)	
2	Set mystaticnm (msn)	
3	Set mystaticgw (msg)	
4	Set mystaticdns (msd)	
5	N/A	No secondary DNS on Voter2
6	Set myipmode	
7	N/A	CLI telnet port fixed at 23
8	N/A	CLI usernames fixed at root and guest
9	Set rootpwd (rpw) Set guestpwd (gpw)	
10-13	N/A	No Dynamic DNS support
14	Set tftpaddr (tad)	Not really equivalent
15	N/A	Ethernet duplex is determined automatically
99	Settings save	

6.3 Offline Menu

Voter	Voter2	Note
1	Set local mode (mod)	
2	Set local cwspeed (cws)	
3	Set local precw (prc)	
4	Set local postcw (poc)	
5	N/A	Fixed at “OFFLINE”
6	N/A	Fixed a “ONLINE”
7	Set local id	
8	Set local hangtime (htm)	
9	Set local plfreq (plf)	
10	Set local pllevel (pll)	
11	Set local deemph (dmp)	
99	Settings save	

6.4